Introduction to Arrays

1. What is Array?

An array is a collection of elements of the same data type stored in consecutive memory locations. In Java, arrays are objects that can hold multiple values under a single variable name. Each element in an array can be accessed using an index number, starting from 0.

Key Characteristics of Arrays:

- **Fixed size**: Once created, the size of an array cannot be changed
- **Same data type**: All elements must be of the same type (int, String, double, etc.)
- **Index-based access**: Elements are accessed using index numbers (0, 1, 2, ...)
- **Zero-indexed**: The first element is at index 0

Array Declaration and Initialization

Declaration:

```java
// Method 1: Declare first, then create
int[] numbers;
numbers = new int[5];

// Method 2: Declare and create in one line
int[] numbers = new int[5];

// Method 3: Declare, create, and initialize with values
int[] numbers = {10, 20, 30, 40, 50};
```

Examples:

```java
// Integer array
int[] scores = new int[4];
scores[0] = 85;
scores[1] = 92;
scores[2] = 78;
scores[3] = 96;

// String array
String[] names = {"Alice", "Bob", "Charlie"};

// Double array
double[] prices = {19.99, 25.50, 12.75, 8.99};
```

Accessing Array Elements

```java
int[] numbers = {10, 20, 30, 40, 50};

// Accessing elements
System.out.println(numbers[0]); // Output: 10
System.out.println(numbers[2]); // Output: 30

// Modifying elements
numbers[1] = 25;
System.out.println(numbers[1]); // Output: 25

// Getting array length
System.out.println(numbers.length); // Output: 5
```

Looping Through Arrays

```java
int[] numbers = {10, 20, 30, 40, 50};

// Using for loop
for(int i = 0; i < numbers.length; i++) {
    System.out.println(numbers[i]);
}

// Using enhanced for loop (for-each)
for(int num : numbers) {
    System.out.println(num);
}
```

2. Importance of Arrays

Arrays are fundamental data structures in programming and offer several important benefits:

Memory Efficiency

Arrays store elements in consecutive memory locations, making them memory-efficient. This allows for faster access to elements compared to other data structures.

```java
// Efficient storage of student grades
int[] grades = new int[100]; // Stores 100 grades efficiently
```

Easy Data Organization

Arrays help organize related data under a single variable name, making code more readable and manageable.

```java
// Instead of creating separate variables:
// String student1 = "John";
// String student2 = "Mary";
// String student3 = "Peter";

// Use an array:
String[] students = {"John", "Mary", "Peter"};
```

Quick Access to Elements

Arrays provide constant-time access to any element using its index, making data retrieval very fast.

```java
int[] productIds = {101, 102, 103, 104, 105};
// Direct access to any product ID
int thirdProduct = productIds[2]; // O(1) time complexity
```

Efficient Processing with Loops

Arrays work seamlessly with loops, enabling efficient processing of large datasets.

```java
int[] sales = {1500, 2300, 1800, 2100, 1950};
int total = 0;

// Calculate total sales efficiently
for(int sale : sales) {
    total += sale;
}
System.out.println("Total Sales: " + total);
```

Foundation for Advanced Data Structures

Arrays serve as building blocks for more complex data structures like lists, stacks, queues, and matrices.

```java
// 2D array (matrix) example
int[][] matrix = {
    {1, 2, 3},
    {4, 5, 6},
    {7, 8, 9}
};
```

Real-World Applications

Arrays are used in various real-world scenarios:

```java
// Storing temperatures for a week
double[] weeklyTemperatures = new double[7];

// Storing student test scores
int[] testScores = new int[30];

// Storing product prices in a shopping cart
double[] cartPrices = {15.99, 23.50, 8.75, 12.00};
```

Quiz Questions

1. What is the index of the first element in a Java array? a) 1 b) 0 c) -1 d) It depends on the array size

Answer: b) 0

2. Which of the following correctly declares and initializes an integer array? a) `int[] numbers = new int[];` b) `int numbers[] = {1, 2, 3};` c) `int numbers = new int[3];` d) `array int numbers[3];`

Answer: b) int numbers[] = {1, 2, 3};

3. What happens if you try to access an array element beyond its size? a) Returns null b) Returns 0 c) Throws ArrayIndexOutOfBoundsException d) Creates a new element

Answer: c) Throws ArrayIndexOutOfBoundsException

4. How do you find the length of an array named 'data'? a) `data.size()` b) `data.length()` c) `data.length` d) `data.count`

Answer: c) data.length

5. Which loop is best suited for traversing all elements of an array when you don't need the index? a) while loop b) do-while loop c) for loop d) enhanced for loop (for-each)

Answer: d) enhanced for loop (for-each)

Practice Coding Questions

1. Create an integer array of size 5, initialize it with values 10, 20, 30, 40, 50, and print all elements using a for loop.

2. Write a program to find the maximum value in an array of integers: `{45, 23, 67, 89, 12, 56}`.

3. Create a String array containing names of 4 cities. Write code to print each city name along with its position in the array (starting from 1).

4. Write a program to calculate and display the average of all elements in a double array: `{85.5, 92.0, 78.5, 96.0, 88.5}`**.**

5. Create an array of 6 integers, take input from user to fill the array, and then display all even numbers from the array.